

AWS S3 Cross Origin Resource Sharing: -

Summary: -

In this lab, I created two S3 buckets and configured Cross-Origin Resource Sharing (CORS) for static website hosting. I uploaded website files, made the buckets public, added bucket policies, enabled static website hosting, and configured CORS permissions between both buckets. Finally, I updated the HTML script to connect the two websites and successfully tested the CORS functionality between them.

Let's Start: -

- Open AWS Console and Search S3.
- Open it and click Create S3 Bucket.

Bucket name [Info](#)

web.cloudservicesdemo.in

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

► **Advanced settings**

ⓘ After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#) [Create bucket](#)

- Now simply scroll down and click Create bucket.

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[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (0)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Name	Type	Last modified	Size	Storage class
No objects				
You don't have any objects in this bucket.				

[Upload](#)

- Now Bucket is created Open it and click on upload.

Upload [Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)

All files and folders in this table will be uploaded.

Find by name

☐ Name ☐ Folder ☐ Type ☐ Size

No files or folders

You have not chosen any files or folders to upload.

- Now click on add files.
- Files provided in resource.
- Then scroll down and click upload.

Files and folders

Configuration

Files and folders (3 total, 2.6 KB)

Find by name

Name	Folder	Type	Size	Status	Error
index.html	-	text/html	488.0 B	✓ Succeeded	-
script.js	-	text/javascript	758.0 B	✓ Succeeded	-
style.css	-	text/css	1.4 KB	✓ Succeeded	-

- Now files is uploaded.
- Close this and go to your created bucket.
- Now as you know that we are creating this bucket to host our static website.
- So we want to make it open so anybody can open this.
- Now by default S3 objects are actually private.
- Nobody can open this and Nobody can access this.
- I want to make it available for general public.
- So I'm going to make my bucket public.

web.cloudservicesdemo.in [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. [Learn more about How IAM analyzer findings work](#)

[View analyzer for ap-south-1](#)

Block public access (bucket settings)

Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access

✓ On

► Individual Block Public Access settings for this bucket

- Go to permissions.

- Then click Edit.

Edit Block public access (bucket settings) [Info](#)

Block public access (bucket settings)

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☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through **new** access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through **any** access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through **new** public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through **any** public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

[Cancel](#)

[Save changes](#)

- Untick Block all public access.
- And click Save changes.

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

[Edit](#)

[Delete](#)

No policy to display.

[Copy](#)

- Now scroll down and click Edit Bucket policy.

Bucket policy

[Policy examples](#)

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

[arn:aws:s3:::web.cloudservicesdemo.in](#)

Policy

1

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

- Now click Add new statement.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Principal": {},
7       "Effect": "Allow",
8       "Action": [],
9       "Resource": []
10    }
11  ]
12 }
```

Edit statement Remove
Statement1

Add actions

Choose a service

Q s3 X

Available

- S3
- S3 Express
- S3 Files
- S3 Object Lambda
- S3 Outposts
- S3 Tables
- S3 Vectors

- Now search S3 in Choose a service.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Principal": {},
7       "Effect": "Allow",
8       "Action": [
9         "s3:GetObject"
10      ],
11       "Resource": []
12    }
13  ]
14 }
```

Edit statement Remove
Statement1

Add actions

[All services](#) > S3

Q getob X

Access level - list

Access level - read

- ☒ [GetObject Info](#)
- ☐ [GetObjectAcl Info](#)
- ☐ [GetObjectAttributes Info](#)
- ☐ [GetObjectLegalHold Info](#)

- Now step2 search GetObject and select it.

Bucket ARN

arn:aws:s3:::web.cloudservicesdemo.in

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Principal": "*",
7       "Effect": "Allow",
8       "Action": [
9         "s3:GetObject"
10      ],
11       "Resource": "arn:aws:s3:::web.cloudservicesdemo.in/*"
12     }
13   ]
14 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

- Now in Principle gives “*”
- Then in resource “copy arn from top and paste in quote”.

[+ Add new statement](#)

JSON Ln 14, Col 1

Security: 0 Errors: 0 Warnings: 0 Suggestions: 0

[Preview external access](#)

Cancel [Save changes](#)

- Now simply scroll down and click Save changes.

web.cloudservicesdemo.in

Info

Objects Metadata **Properties** Permissions Metrics Management File systems - new Access Points

Bucket overview

AWS Region Asia Pacific (Mumbai) ap-south-1	Amazon Resource Name (ARN) arn:aws:s3:::web.cloudservicesdemo.in	Creation date 🕒
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- Now go to properties section.

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

[Edit](#)

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

[Create Amplify app](#)

S3 static website hosting
Disabled

- Now scroll down and click Edit in Static website hosting.

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

- ☐ Disable
- ☒ Enable

- Enable it.

Index document

Specify the home or default page of the website.

index.html

Error document - optional

This is returned when an error occurs.

error.html

Redirection rules – optional

Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

- Scroll down and index document enter your html page name with extension.

JSON Ln 1, Col 1 ✖ Errors: 0 ⚠ Warnings: 0 ⚙

Cancel

Save changes

- scroll down and click save changes.

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

ⓘ We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting

Enabled

Hosting type

Bucket hosting

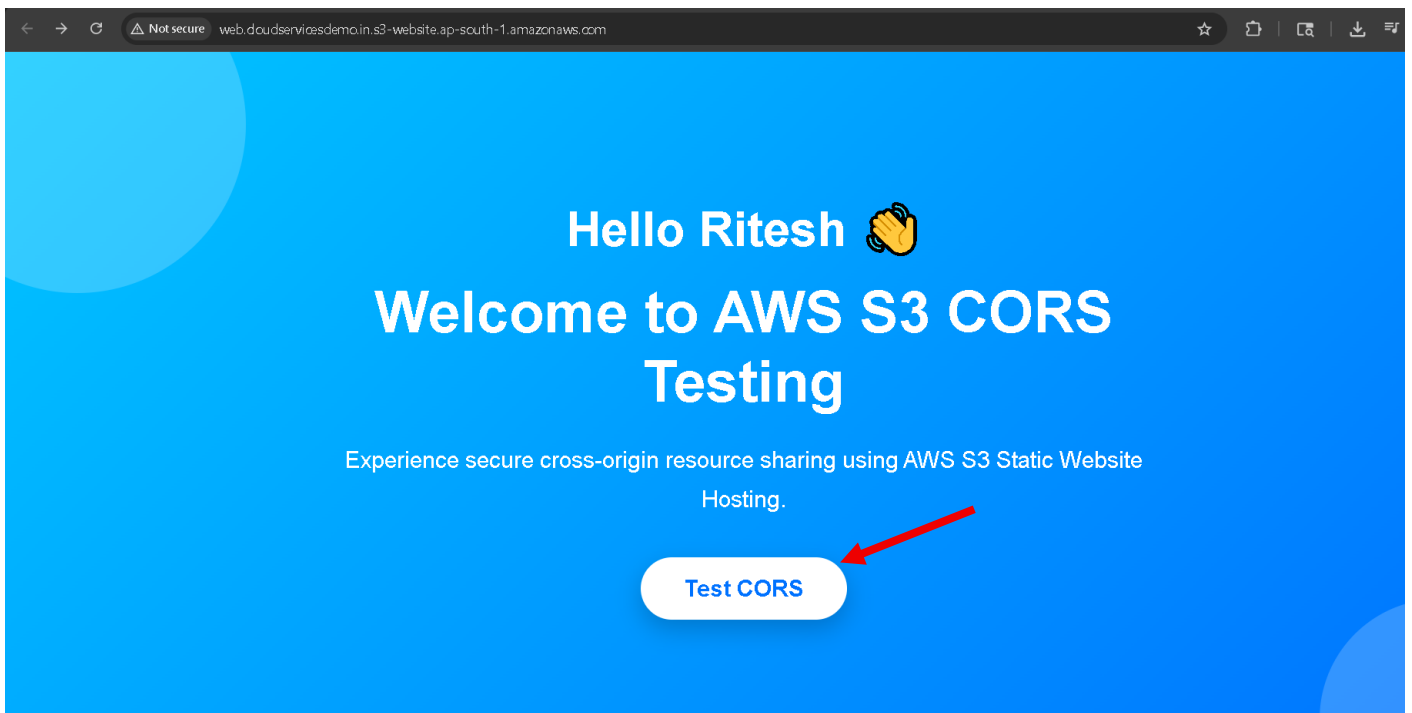
Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://web.cloudservicesdemo.in.s3-website-ap-south-1.amazonaws.com>

Edit

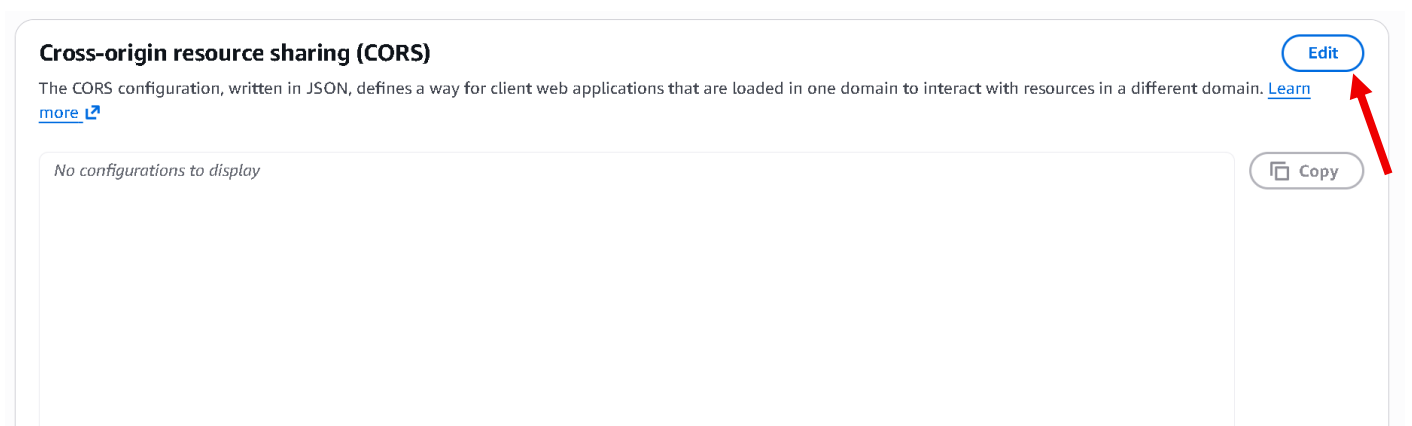
- now scroll down and see our website is hosted.
- Copy bucket website endpoint and open in browser new tab.



- It's working now when we click it will redirect to another bucket.



- If I click now you see 404 Not Found.
- So we need to create one more bucket.
- And perform same operation that we perform in starting bucket creation.



- Now go inside new bucket permission section and performing all previous steps scroll down and in Cross-origin resource sharing click Edit.

Cross-origin resource sharing (CORS)

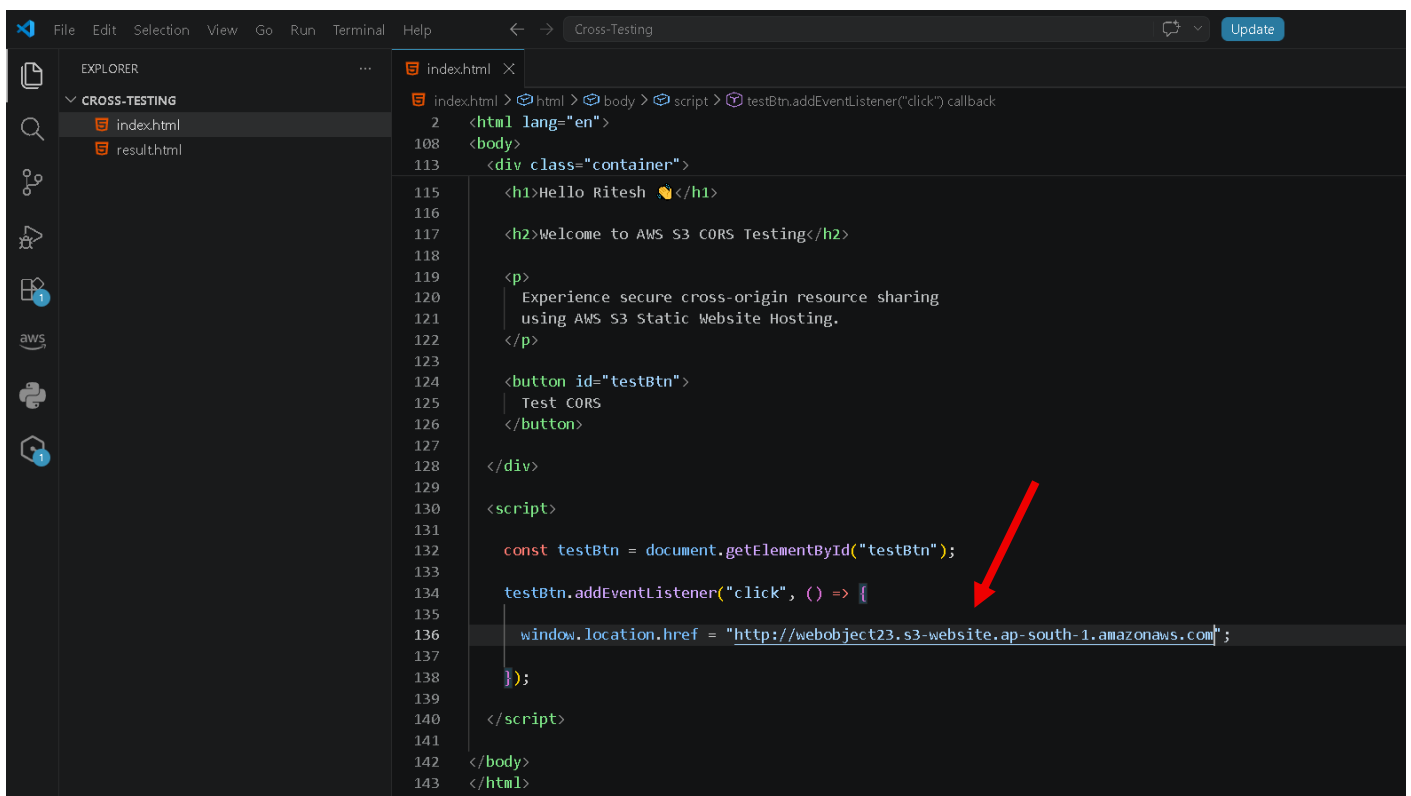
The CORS configuration, written in JSON, defines a way for client web applications that are loaded in one domain to interact with resources in a different domain. [Learn more](#)

```
1 [
2   {
3     "AllowedHeaders": ["*"],
4     "AllowedMethods": ["GET"],
5     "AllowedOrigins": [
6       "http://web.cloudservicesdemo.in.s3-website.ap-south-1.amazonaws.com"
7     ]
8   }
9 ]
```



```
[
{
  "AllowedHeaders": ["*"],
  "AllowedMethods": ["GET"],
  "AllowedOrigins": [
    "First Bucket Static URL Paste"
  ]
}
]
```

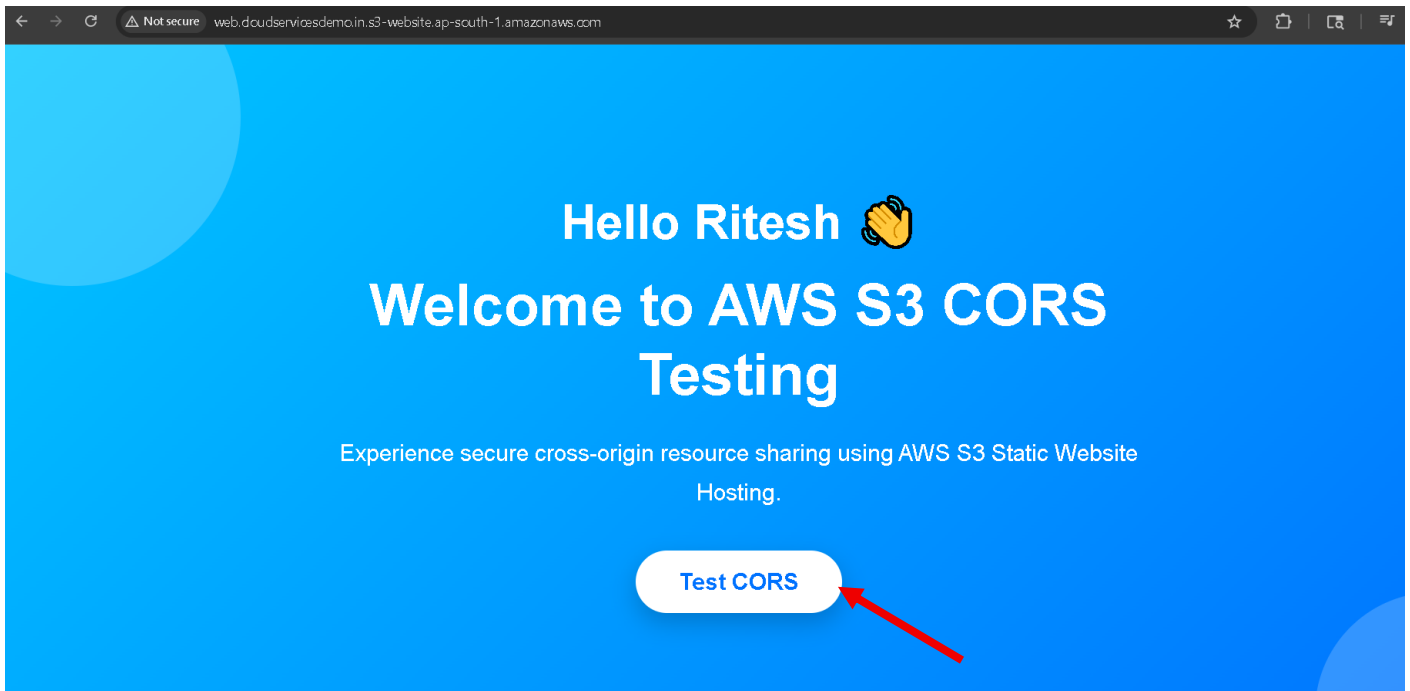
- Now simply scroll down and click Save changes.
- Now after this go to properties of new Bucket and scroll down to end and like previously enable the Static web hosting.
- Now all our work is done.
- But we need to change our 1st Bucket HTML File.
- Now open that html file in your VSCode.



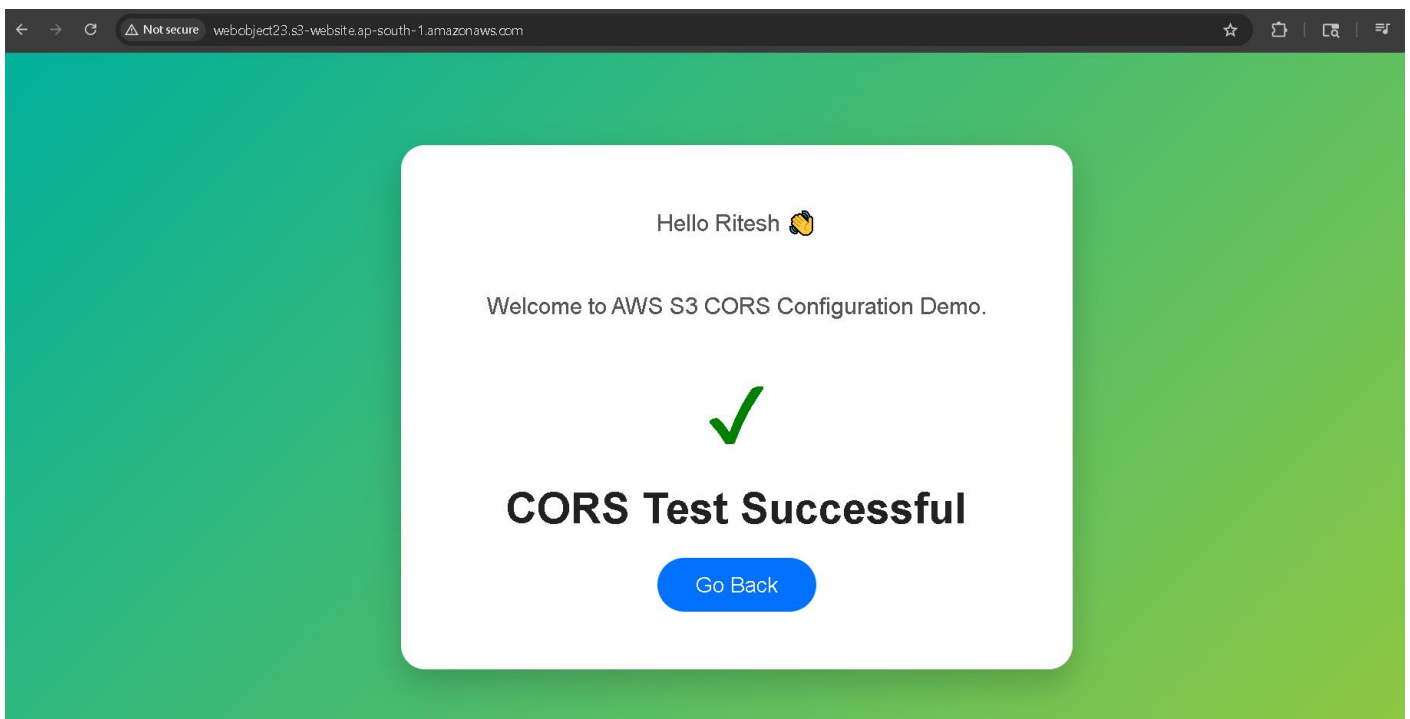
```
File Edit Selection View Go Run Terminal Help
Cross-Testing
index.html X
index.html > html > body > script > testBtn.addEventListener("click") callback
2 <html lang="en">
108 <body>
113 <div class="container">
115 <h1>Hello Ritesh </h1>
116
117 <h2>Welcome to AWS S3 CORS Testing</h2>
118
119 <p>
120 | Experience secure cross-origin resource sharing
121 | using AWS S3 Static Website Hosting.
122 </p>
123
124 <button id="testBtn">
125 | Test CORS
126 </button>
127
128 </div>
129
130 <script>
131
132 const testBtn = document.getElementById("testBtn");
133
134 testBtn.addEventListener("click", () => {
135
136   window.location.href = "http://webobject23.s3-website.ap-south-1.amazonaws.com";
137
138 });
139
140 </script>
141
142 </body>
143 </html>
```

- Now go to script and in windows location.href = "Paste 2nd bucket Script hosting URL)
- Go to first bucket delete uploaded files and upload this new html file.

- Then go to properties of 1st bucket and scroll down and copy URL From Static hosting and open in new tab.



- It's working now click Test CORS.



- Now its working.